



Comprehensive evaluation of machine learning models for real-world air quality prediction and health risk assessment by AirQ+

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Abstract

This study extensively examines five distinct machine learning models used to predict hourly air particulate matter concentrations. The study used real-world data, including pollutant levels and various meteorological parameters, for model training and evaluation, making the study more reliable and effective. The study focused on capturing short-term trends in pollutant concentrations and meteorological conditions. Results showed varied model performances. The Ridge Regression model exhibited a moderate R^2 value of 0.44 for $PM_{2.5}$ prediction and an impressive R^2 of 0.91 for PM_{10} prediction. Support Vector Regression showed strength in $PM_{2.5}$ prediction ($R^2 = 0.83$) but faced challenges in forecasting PM_{10} . Random Forest and Extra Trees Regression demonstrated robust overall performance, particularly in PM_{10} forecasting ($R^2 = 0.75$). Extreme Gradient Boosting displayed competitive results for both $PM_{2.5}$ and PM_{10} ($R^2 = 0.80$ and 0.81). Each model's identified strengths and limitations provide valuable insights for air quality management, offering a foundation for future research and the development of machine learning models in the continuous pursuit of accurate and timely air quality predictions. The AirQ+ model was used to estimate the health effects of $PM_{2.5}$ exposure and predict the long-term mortality rates associated with $PM_{2.5}$. The average estimated attributable proportion for all years is 10.2% (with a range of 6.5% to 13.2%). The results show differing trends in estimated mortality rates, underscoring the need for targeted interventions to reduce the public health risks associated with exposure to polluted air.

Keywords Air quality · Predictive modelling · Machine learning · Regression · Random forest · Health risk assessment

Introduction

Air pollution significantly impacts human health and the environment, with air pollutants posing a significant concern for both aspects. Particulate matter is highlighted as a hazardous pollutant due to its ease of inhalation and potential to harm the respiratory system (Dimitriou et al. 2022). Research studies have established a direct and significant correlation between poor air quality and numerous health complications (Guo et al. 2024). Exposure to poor air quality has been linked to a range of illnesses, including lung cancer, respiratory infections, stroke, and chronic obstructive pulmonary disease (Bala et al. 2021). According to recent

estimates, ambient air pollution is responsible for the premature deaths of 4.2 million individuals worldwide in 2019 (WHO 2022). Therefore, predicting $PM_{2.5}$ and PM_{10} concentrations has garnered global attention from governments, researchers, and the public.

Air quality has become a significant concern in Turkey due to various factors that contribute to environmental degradation. The increase in population density, building density, and energy consumption has deteriorated outdoor air quality in urban regions. Recent research on air quality in Turkey has provided valuable insights. Studies have shown a clear link between meteorological conditions and air pollution levels, particularly during winter in cities, where rapid urbanization and increased energy consumption have contributed to declining outdoor air quality (Akpınar et al. 2008). Natural gas usage has also significantly affected air quality, particularly sulfur dioxide concentrations.

Furthermore, this research has highlighted the connection between air pollution and health outcomes, including the impact of air pollution on emergency department visits

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and hospitalizations among people with COPD in northern Turkey (Sağır and Bal Özkaptan 2022). The COVID-19 pandemic has further emphasized the importance of understanding the interplay between meteorological conditions, air pollution, and virus transmission in various cities (Yılmaz et al. 2023). These findings underline the urgent need for comprehensive air quality monitoring, strict regulations, and policy interventions to address the environmental and public health challenges of air pollution in Turkey.

Air quality monitoring networks are effective methods to assess ambient air quality. These stations are guided by objectives outlined in the ambient air quality directive, which outlines the criteria and procedures for evaluating air quality in member states (Qi et al. 2018). However, expanding the coverage of such monitoring networks poses challenges related to factors like population density, geographical location, costs, and the maintenance life cycle of monitoring devices. The limitation of public administration budgets restricts the feasibility of increasing the number of monitoring stations (Fang et al. 2019). By developing accurate and reliable air quality prediction models, it becomes possible to overcome the constraints of physical monitoring networks. Here's why air quality prediction is essential in this context: (1) Predictive models allow for the efficient allocation of monitoring resources. Instead of deploying sensors across vast areas, predictive models enable the identification of critical locations where air quality issues are more likely to occur, optimizing the use of limited resources. (2) Air quality prediction models provide the ability to forecast pollution events in advance. This early warning system allows authorities to take timely and targeted interventions to mitigate the impact of poor air quality, safeguarding public health. (3) Predictive modeling can be a cost-effective alternative to establishing and maintaining an extensive physical monitoring network. By relying on mathematical algorithms and data analytics, it is possible to extrapolate air quality information for areas without direct monitoring stations (Brown et al. 2024). (4) Accurate air quality predictions inform policy planning and decision-making. Governments and regulatory bodies can use this information to formulate effective strategies for pollution control, emissions reduction, and urban planning. (5) Air quality prediction models contribute valuable data for scientific research on atmospheric processes, pollutant dispersion, and the impact of human activities on air quality (Dubey et al. 2019). This knowledge can refine predictive models and improve our understanding of air quality dynamics. In conclusion, air quality prediction is pivotal in overcoming the challenges of expanding monitoring networks. By harnessing data analytics and modeling, a more comprehensive and cost-effective assessment of ambient air quality can be achieved, ultimately supporting efforts to protect public health and the environment.

To address the difficulties linked to the expansion of air quality monitoring networks, this paper aims to predict hourly $PM_{2.5}$ and PM_{10} concentrations in Aksaray, Türkiye, and to investigate the effects of critical determinants (meteorological and emission) on prediction accuracy. To overcome the abovementioned constraint, Machine Learning (ML) has emerged as a promising solution. ML techniques offer the potential to develop robust models that can establish specific relationships between data collected from existing monitoring station and predict pollutant concentrations, thereby enhancing the efficiency and effectiveness of air quality monitoring efforts (Rybarczyk and Zalakeviciute 2018).

The AirQ+ software, developed by the World Health Organization (WHO) Regional Office, is a powerful tool to estimate the health impacts of air pollution. It has been extensively utilized in recent studies to predict the potential effects of air pollution on human health, providing valuable insights into the environmental burden of diseases such as acute lower respiratory tract disease (ALRI), chronic obstructive pulmonary disease (COPD), and ischemic heart disease (IHD). Additionally, the software has proved effective in accurately determining mortality rates caused by exposure to various atmospheric pollutants and evaluating the health benefits of reducing particulate matter (PM) levels in densely populated urban areas.

Motivation

The first goal of this manuscript is to address the increasing environmental challenges caused by air pollution, particularly PM_{10} and $PM_{2.5}$ levels in the Aksaray region of Türkiye. As the area continues to urbanize and industrialize, it has become crucial to address air quality concerns. The summary of the studies conducting air pollution predictions is depicted in Table S1. This research draws inspiration from machine learning techniques used by researchers such as Hu et al. (Hu et al. 2017) and Grange et al. (Grange et al. 2018) utilized random forest models for air pollution prediction. This research also builds on the approach taken by Bellinger et al. (Bellinger et al. 2017), Bai et al. (Bai et al. 2018), Nemade (Nemade 2019) and Qadeer et al. (Qadeer et al. 2021) emphasizes the importance of data mining, machine learning algorithms, and air pollution forecasting models. The aim is to develop specific air pollution prediction models that are precise and context-specific, with a particular focus on PM_{10} and $PM_{2.5}$ levels in Aksaray, Türkiye. To achieve this, pivotal variables such as air quality measurements and meteorological conditions have been integrated to enhance the accuracy of the predictive model. The urgency of this research is highlighted by studies such as those by Sharma et al. (Sharma et al. 2022) and Tran et al. (Tran et al. 2023), which demonstrates the adverse health effects

of air pollution. The second goal is to evaluate the impact of air pollution on the long-term health of the population in Aksaray; the AirQ+ software has been used to assess the health effects. It is crucial to develop strategies to protect the citizens from the harmful effects of air pollution, as this will promote public health in the long run.

Research gap

The proposed predictive model is a valuable tool that can facilitate the implementation of timely regulatory measures to mitigate these adverse health effects. The ultimate objective is for this research to contribute significantly to the advancement of environmental sustainability and public well-being, particularly in response to the escalating challenges posed by air quality in the Aksaray region.

There are still some gaps in the current literature regarding predicting $PM_{2.5}$ and PM_{10} concentrations and the health impact of $PM_{2.5}$ in the Aksaray region of Türkiye. Specifically, there is a need for more research on using multiple machine learning (ML) models for short-term forecasting. Here are the gaps that have been identified:

1. *Limited Comparative Analysis of Multiple ML Models:* The existing studies on predicting $PM_{2.5}$ and PM_{10} concentrations in Aksaray need comprehensive comparative analysis of various ML models. More research is required to compare and contrast distinct ML algorithms' performance systematically. Such comparisons are essential to identify the most effective model or combination of models tailored to Aksaray's intricate air quality dynamics.
2. *Absence of Short-Term Forecasting Focus:* Most current research on air pollution prediction in Türkiye focuses on broader aspects, with limited emphasis on short-term forecasting over a 24-hour horizon. Given the dynamic nature of atmospheric conditions and emission patterns, studies geared explicitly toward short-term predictions are needed. This will enable a more timely response to potential air quality fluctuations in the region.
3. *Insufficient Integration of Actual Data Comparison:* Many studies need to comprehensively compare the predictions made by ML models and the actual PM_{10} and $PM_{2.5}$ concentrations observed in the subsequent 24 hours. A comparative analysis is necessary for the validation and reliability assessment of the predictive models, making it crucial to bridge this gap for a more comprehensive understanding of their real-world applicability.
4. *Lack of Health Impact Studies on the Region:* The literature needs comprehensive health impact studies specific

to Aksaray. This study utilized advanced tools such as AirQ+ to conduct a thorough health impact assessment.

This study aims to improve the accuracy of predicting air pollution caused by $PM_{2.5}$ and PM_{10} in Aksaray, Türkiye. It uses a multifaceted machine learning approach with various algorithms such as the Ridge Model, Support Vector Regression (SVR), Random Forest Regression (RFR), Extra Trees Regression (ETR), and Extreme Gradient Boosting (XGBoost) to contribute to the field significantly. The study also considers incomplete data using the missForest method to minimize the impact of missing values. John Tukey's method is applied to identify and remove outliers, ensuring a comprehensive and robust data analysis. Moreover, the study uses the Boruta algorithm for feature selection and examines the impact of meteorological parameters on air quality predictions through two different scenarios. These aspects provide a more nuanced understanding of the factors contributing to air pollution in the area.

Additionally, the study predicts concentrations for both $PM_{2.5}$ and PM_{10} over the next 24 hours, facilitating short-term air quality forecasting and timely decision-making. The evaluation metrics used, including the coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE), ensure a comprehensive understanding of model performance. Finally, the study contextualizes its findings in Aksaray, Türkiye, to provide valuable insights that can inform localized strategies for air quality management. The meticulous application of diverse machine learning methods, coupled with the strategic handling of incomplete data, hourly prediction, and rigorous evaluation metrics, makes this study a significant contribution to the field of air pollution prediction. It is expected to have implications for academia and practical applications, particularly in environmental management and public health in Aksaray, Türkiye, and beyond. In addition, this study includes a thorough analysis of the health effects of exposure to $PM_{2.5}$ in Aksaray. Advanced health risk assessment tools such as AirQ+ quantify the potential health impacts of both short-term and long-term exposure to particulate matter. By examining pollution levels in relation to health data, the research provides insights into the prevalence of mortality rates linked to air pollution. This emphasizes the need for effective air quality management and underscores the public health benefits of lowering $PM_{2.5}$ concentrations. The findings are intended to inform local health policies and contribute to developing targeted interventions to safeguard vulnerable populations in Aksaray.

Material and methods

Proposed workflow for air pollution level predictions

As previously mentioned, this research aims to improve the accuracy of air pollution level predictions, focusing on $PM_{2.5}$ and PM_{10} in an urban area. Fig. 1 represents the workflow of the proposed architecture for predicting $PM_{2.5}$ and PM_{10} concentration. The data for this study was taken from the National Air Quality Monitoring Network and the Turkish State Meteorological Service. Following the data acquisition process, a preliminary analysis of the complete dataset was conducted to detect any outliers and missing values. This step was taken to ensure the data was accurate and reliable for further analysis. As a post-analysis step, the study conducted an in-depth investigation into the correlation between pertinent features and pollutant concentration. The objective was to understand the underlying dynamics better and identify possible causal relationships. Then, the data has been partitioned into distinct training and testing subsets. After that, a comprehensive machine learning approach was employed, incorporating various algorithms such as Ridge Model, SVR, RFR, ETR, and XGBoost. The impact of meteorological parameters on air quality predictions was also examined through two distinct scenarios. These methodological intricacies provide a nuanced understanding of the factors influencing air pollution levels in the research area. In the end, concentrations for both $PM_{2.5}$ and PM_{10} for the next 24 hours were

forecasted, which can help facilitate short-term air quality forecasting and timely decision-making. The performance of the proposed methods was evaluated by applying various performance metrics such as R^2 , MAE, and RMSE.

Study area and data set

To address concerns related to air quality, Türkiye has taken a proactive approach by setting up an extensive network of ambient air quality monitoring stations (AQMs) throughout the country. The National Air Quality Monitoring Network, established in 2005, is a leading initiative to improve air quality by providing reliable and representative air quality data. The network collaborates with strategically positioned monitoring stations nationwide, which gather time series measurements detailing ambient pollutant concentrations. The hourly averages of these measurements are used to assess air quality dynamics. This data helps authorities identify pollution hotspots and make informed decisions to improve air quality. For this study, an AQM located in Aksaray, at coordinates $38^{\circ}21'52.1''N$ and $34^{\circ}01'42.7''E$, has been selected as an urban monitoring station (Fig. 2). The station is 970 meters above sea level, 60 meters from the nearest road, and approximately 3 kilometers from the industrial zone. It is operated and managed by the Ministry of Environment, Urbanization, and Climate Change of Türkiye and records hourly concentrations of air pollutants, including PM_{10} , $PM_{2.5}$, SO_2 , CO , and O_3 . This information is used to assess the air quality in the region and take appropriate measures to improve it. In addition to the AQM, a meteorological station located in the city center ($38^{\circ}22'13.8''N$ and

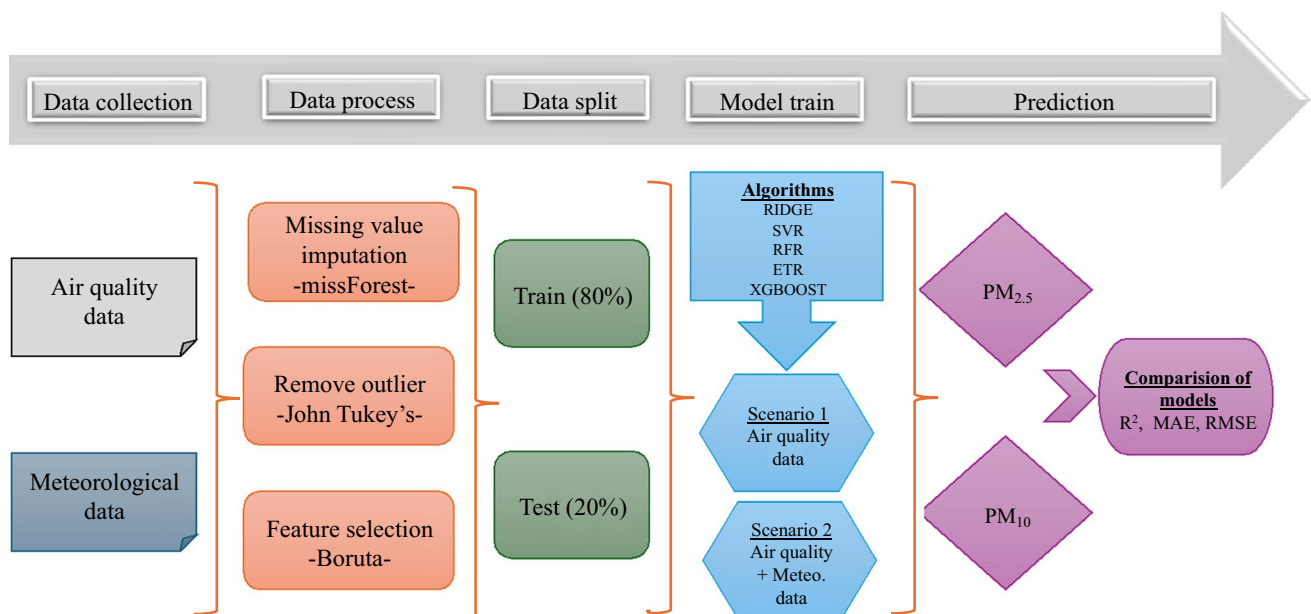


Fig. 1 $PM_{2.5}$ and PM_{10} concentration prediction methodology: workflow framework

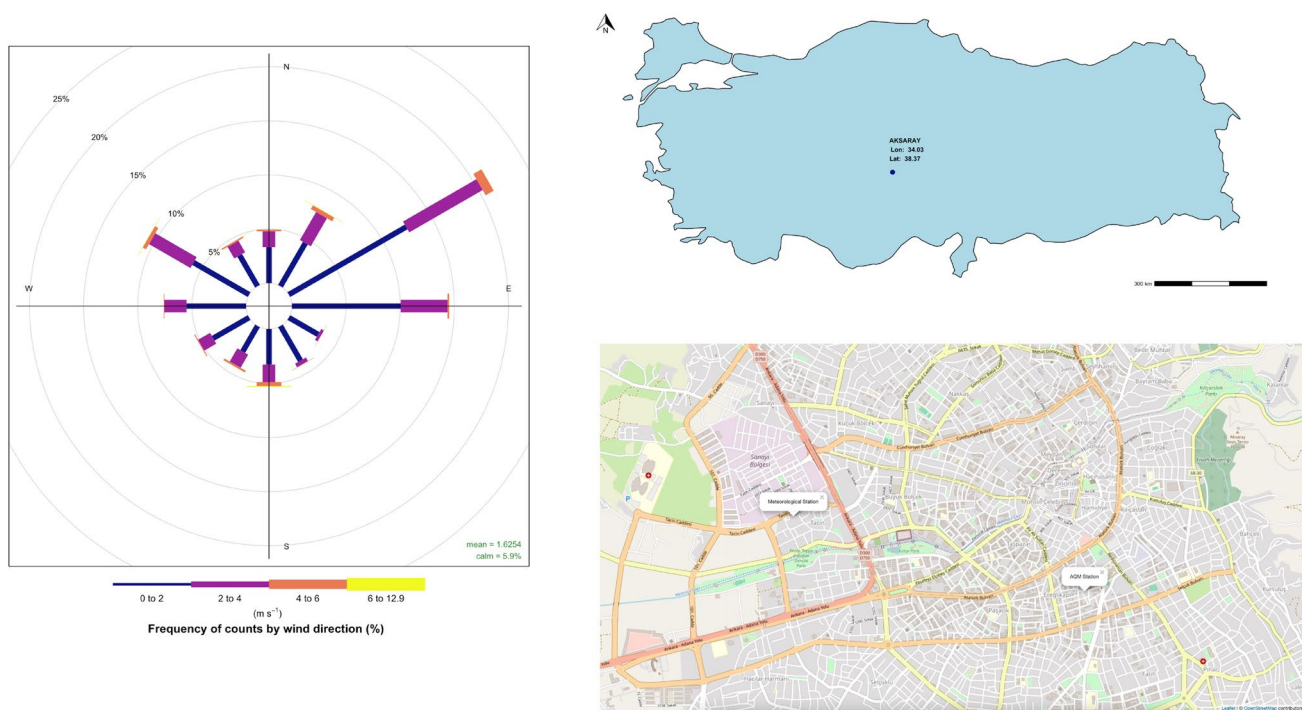


Fig. 2 Map of the study area i.e., Air quality monitoring station and meteorology station, Aksaray, Türkiye

33°59'55.3"E) measures various meteorological parameters such as wind speed, wind direction, temperature, humidity, precipitation, atmospheric pressure, soil temperature, and snow depth.

This study has used hourly measurements of air pollutants such as PM_{10} , $PM_{2.5}$, SO_2 , and CO, along with meteorological observations, including wind direction, wind speed, temperature, and humidity, collected between 2019 and 2022. This has resulted in a dataset comprising 35,064 rows of data. The hourly statistics of the raw data set are shown in Table S2. This number has been calculated considering 24 hours per day, 365 days per year, and four years, including one leap year (2020). The study aims to analyze this data and understand the relationship between air quality and meteorological conditions. Moreover, the study seeks to predict $PM_{2.5}$ and PM_{10} levels.

Data processing

Data preparation is a crucial step in air quality prediction studies, as it impacts the accuracy and reliability of predictive models. One of the standard methods used in data preparation is advanced techniques for handling missing data. Research has shown that missing values in air quality datasets can significantly impact the quality of predictions and decision-making processes (Rad et al. 2024; Imam

et al. 2024). Techniques such as the random forest algorithm can effectively handle missing data in air quality datasets, enabling researchers to explore and analyze the data more comprehensively (2023; Koçak 2024). Outliers in air quality datasets are also crucial in the accuracy and reliability of prediction and modeling efforts. Outliers can significantly skew results and lead to inaccurate forecasts, impacting the effectiveness of air quality control measures and environmental management decisions. Therefore, identifying and removing outliers is essential to ensure the quality and integrity of air quality data for predictive modeling. Studies such as Chen et al. (Chen et al. 2021) and Wang et al. (Wang et al. 2020) emphasizes the importance of creating accurate air quality prediction models to support meteorological forecasts and air quality control. These models are prone to overfitting when outliers are present, highlighting the necessity of outlier removal for robust predictions. Another crucial aspect of data preparation is feature selection, which can help to enhance the performance and interpretability of predictive models. Algorithms can be utilized to identify the most relevant features for modeling air quality (Kapadia and Jariwala 2021). By carefully addressing missing values, removing outliers, and selecting pertinent features, researchers can ensure that the data used for air quality prediction is robust and optimized for accurate modeling and forecasting.

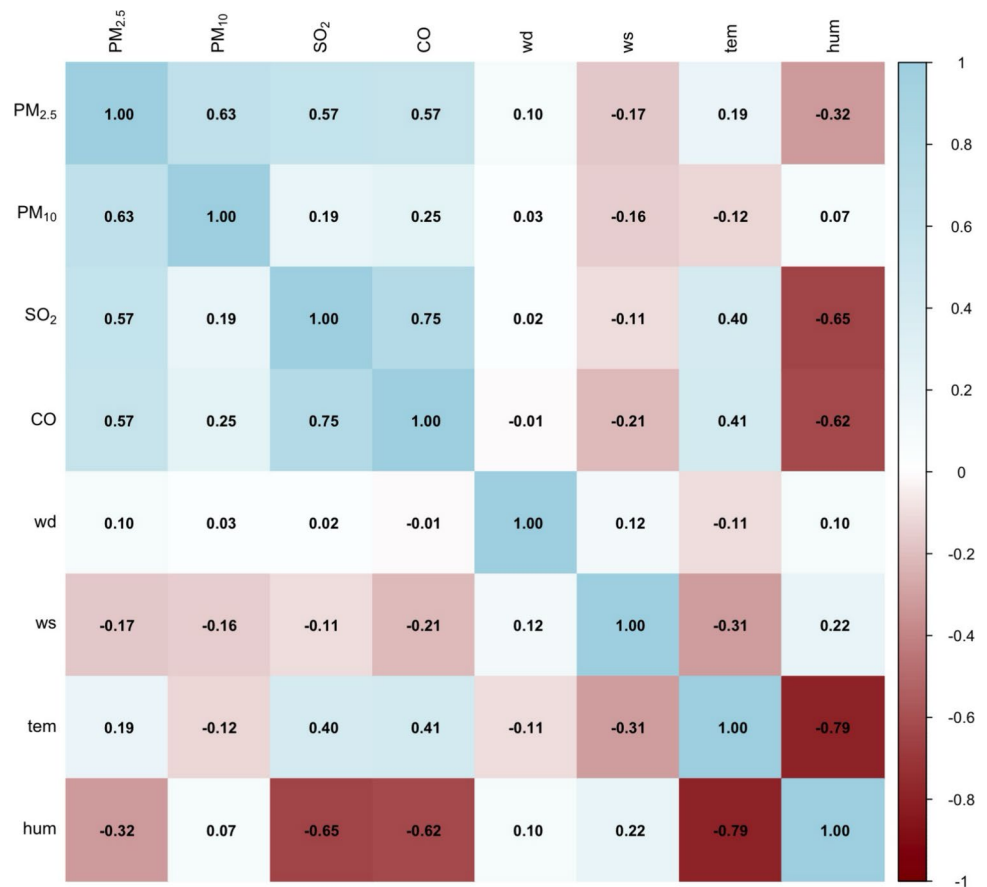
This study uses the missForest algorithm (RStudio®) to calculate missing value imputation. MissForest is a non-parametric method specially designed for mixed-type data and can handle missing values in variables of various types, including continuous and categorical variables. This algorithm has outperformed other standard imputation methods, making it a reliable choice for imputing missing data (Junninen et al. 2004). It has been used as a benchmark for non-parametric imputation methods due to its favorable performance. The missForest algorithm was implemented in this study, and a four-step process was followed (Ahmat Zainuri et al. 2015). To deal with missing observations in a variable, they were replaced with the mean value of that variable. During the Random Forest (RF) model training process, the variable containing imputed missing data was designated as the target variable (dependent variable). The other variables were used to predict this target variable, and the RF model was trained using the complete, non-missing data of the target variable. Then, the missing entries were substituted with estimated values derived from the learned model. The different variables were assigned as predictors to construct the RF model, and step 2 was repeated for each variable with missing entries. The initial imputation iteration was considered complete once RF models for every variable with missing data were trained. Subsequently, steps 2 and 3 were repeated until the stopping conditions were met. The stopping conditions were based on the trained RF models' mean square error (MSE). The imputation procedure was concluded when the MSE of iteration (i) exceeded the MSE of the previous iterations, i.e., ($i-1$). At this point, the results were those obtained from the last iteration.

As employed in this study, an efficient way to identify outliers is to use John Tukey's method, a statistical technique widely used in outlier detection. This method uses Tukey's fences, a reliable statistical technique for outlier detection. Tukey's method sets boundaries based on the interquartile range, effectively identifying data points outside these boundaries as outliers (Spoel et al. 2019). This method calculates a sample's interquartile range (IQR) and three fundamental components: r , $Q1$, and $Q3$. The parameter r is a multiplier of the IQR that determines the boundaries for identifying outliers. Specifically, any data point that falls below ($Q1 - r \times IQR$) or above ($Q3 + r \times IQR$) is considered an outlier. $Q1$ and $Q3$ represent the first and third quartiles of the data set, respectively (Narkhede et al. 2023). These flagged data points can be further analyzed to understand their significance better. In this study, the parameter r was set to 1.5 in accordance with Tukey's conventional method. Outlier thresholds were calculated separately for each variable using the $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ formulas. For variables with physical constraints—such as pollutant concentrations, wind speed, and humidity—negative lower bounds were adjusted to zero. The numerical thresholds

derived for all key variables are presented in Table S4. Outliers identified through this method were excluded from the dataset prior to model training in order to minimize skewness and avoid potential overfitting.

This study uses the Boruta algorithm (RStudio®) to select the most relevant features in a dataset. The Boruta algorithm is a technique for choosing the most pertinent features of a dataset. It is based on Random Forest models and is known for its ability to measure the importance of features within the classifiers (Ahmed et al. 2021). By selecting the most critical features, Boruta can enhance the accuracy and efficiency of predictive models. This study used different machine-learning techniques to predict hourly $PM_{2.5}$ and PM_{10} concentrations. One crucial aspect of constructing a machine learning model is selecting the right inputs. Including noisy or irrelevant inputs can negatively impact the training process and result in a complex model with limited generalization capabilities. To address this challenge, an input selection algorithm compares the importance of original attributes with their permuted duplicates. The algorithm performs a top-down search for significant characteristics and excludes irrelevant features from the model. The analysis presented in Fig. S1 demonstrates that the Boruta algorithm has confirmed the significance of all input parameters in predicting both $PM_{2.5}$ and PM_{10} levels. The Boruta algorithm assesses feature importance by comparing the original variables against permuted shadow features, thereby excluding any variables that do not substantially outperform their randomized counterparts. In this instance, no variables were excluded, indicating that each contributed meaningfully to the predictive models. This emphasizes the essential role of the selected features in effectively predicting concentrations of air pollutants. By employing the Boruta algorithm for feature selection, the study successfully developed predictive models that are optimized for both performance and generalization.

The Spearman correlation coefficients derived from the dataset provide significant insights into the relationships among various environmental factors (Fig. 3). This approach is commonly employed in air quality research to investigate monotonic associations between meteorological variables and pollutant concentrations (Maniat et al. 2024; Geng et al. 2021). The significance levels (p -values) corresponding to the Spearman correlation coefficients were computed, and only statistically significant correlations ($p < 0.05$) were interpreted in the analysis. The analysis shows a strong positive correlation between $PM_{2.5}$ and PM_{10} concentrations, indicating that these two pollutants increase together and may have a common source or atmospheric conditions. The positive correlations between $PM_{2.5}$, PM_{10} , SO_2 , and CO concentrations suggest that these pollutants share a common origin or simultaneous emission sources. Identifying and addressing these familiar sources can help

Fig. 3 Spearman rank correlation matrix between features

develop targeted pollution control measures. Understanding the interplay between different pollutants is crucial for comprehensive air quality management, as the co-occurrence of these pollutants may increase health risks and environmental impacts. The negative correlation between temperature and PM_{2.5} and PM₁₀ is interesting. Higher temperatures can positively affect reducing particulate matter levels due to increased atmospheric dispersion or enhanced pollutant removal mechanisms during warmer periods. However, it also raises questions about the potential implications of rising global temperatures on air quality dynamics and whether this negative correlation holds in different regions. The negative correlation between humidity and temperature emphasizes the complex relationship between meteorological factors and air quality. Higher temperatures are associated with lower humidity levels, which may impact the formation and suspension of particulate matter. The negative correlation between wind speed and humidity suggests that increased wind speed might contribute to lower humidity levels. This relationship is essential in understanding the role of atmospheric dynamics in shaping air quality. Higher wind speeds could enhance pollutant dispersion, improving air quality in specific scenarios. However, further investigation is needed to discern the interactions between wind patterns, humidity, and pollutant concentrations.

In conclusion, the analysis of Spearman correlation coefficients provides insights into the complex relationships between environmental factors and is valid for informed decision-making in air quality management. These insights contribute to a better understanding of the factors influencing air quality. They are instrumental in developing effective strategies to mitigate the adverse impacts of air pollution on human health and the environment.

Data split

After identifying the features, the following step was to divide the hourly data into training and test data. To achieve this, 80% of the data was allocated for training, while the remaining 20% was set aside for testing. This division is based on previous research studies, where numerous authors suggested utilizing this range. To assess the model's generalization ability, a 5-fold cross-validation method was used. This method involved dividing the training set into five equal parts. Four parts were used each time for training; the remaining part was evaluated as the validation set. This technique helped to improve the model's generalization ability during exercise, which could lead to credible results during testing. Assess the model on one part after training on four parts to identify each model's best set of hyperparameters.

In time-series modeling, it is crucial to uphold the temporal order of observations. To achieve this, 5-fold cross-validation was applied within the training dataset, conducting it in a chronological sequence rather than through random sampling. Each fold represented a continuous time block, thereby ensuring that future observations were not utilized during the training phase. This methodology effectively mitigates the risk of temporal leakage and preserves the integrity of learning from temporal features.

Model training and hyperparameters

After dividing the dataset into training and testing sets, five separate machine learning models were trained using the same training data. The training strategy is illustrated in Fig. 1, which outlines each scenario's specific meteorological and ambient pollutant concentrations, along with their corresponding targets, $PM_{2.5}$ and PM_{10} . Each of the five machine-learning models was fed the same data but was trained using various techniques, such as decision trees, support vector machines, and random forests. The models were trained to predict the values of $PM_{2.5}$ and PM_{10} based on the input data. Table 1 outlines the ML models and hyperparameters employed in the study. The selection of these models was informed by a thorough literature review of prior research that predicted air pollutant concentrations. All the models utilized in the study are readily available in RStudio®. A series of trials determined the best hyperparameter set, with cautious attention to preventing overfitting. To identify the optimal sets of hyperparameters for each model, a grid search methodology combined with 5-fold cross-validation was implemented on the training dataset. Model performance was assessed based on the attainment of the lowest root mean square error (RMSE) and the highest R^2 values during the cross-validation process. This approach effectively mitigated the risk of overestimating model performance due to overfitting. Furthermore, to further reduce overfitting, regularization techniques, such as shrinkage in XGBoost and weight decay in Ridge Regression, as well as sub-sampling strategies, were incorporated into the model configurations. Ultimately, the models were validated using a held-out test set to verify their generalization capabilities.

Health impact assessment with AirQ+

The study examined the intricate relationship between air quality and public health in Aksaray by combining diverse datasets. Health data was collected from the Turkish Statistical Institute (TUIK) and its Health and Social Protection Database, which included various health indicators such as disease prevalence, mortality rates, healthcare usage, and social welfare programs. This comprehensive dataset provided insight into the population's health status, healthcare requirements, and the effectiveness of social protection measures. To gauge the potential impact of air quality on public health, the study obtained total mortality counts from the Address-Based Population Registration System (ABPRS). These counts offered valuable information on recorded deaths, enabling the assessment of health outcomes and mortality trends related to air quality.

The AirQ+ (version 2.2) software tool, developed by the European regional office of the World Health Organization, is utilized to quantify the health effects of $PM_{2.5}$ (WHO 2023). This comprehensive program is designed to estimate the impact of air pollution on public health using established methodologies and concentration-response functions derived from rigorous epidemiological studies. AirQ+ incorporates a range of parameters, including attributable proportion, relative risk, and baseline incidence, to estimate the health impact of air pollution on a specific population. Epidemiological studies have significantly contributed to the development of AirQ+. These functions enable the calculation of attributable proportions and the number of cases related to air pollutant exposure. Key factors such as baseline rates of health outcomes, cut-off values, and relative risk values are essential for quantifying the long-term effects, particularly those of $PM_{2.5}$. AirQ+ utilizes default values for relative risks associated with different health endpoints and employs a linear log method for risk estimation across varying pollutant concentrations. This method involves a logarithmic transformation of the input data to transform a non-linear relationship into a linear one, followed by applying a linear regression model to estimate the relative risks.

To assess the health impacts of $PM_{2.5}$ pollution utilizing AirQ+, baseline health data specific to Aksaray were

Table 1 ML models and parameters

ML Models	Model Hyperparameters
Ridge	Lamda, weight decay
SVR	Sigma, C
RFR	Multiple try (mtry)
ETR	Randomly Selected Predictors, Splitting Rule, Minimal Node Size
XGBOOST	Boosting Iterations, Max Tree Depth, Shrinkage, Subsample Ratio of Columns, Minimum Sum of Instance Weight, Minimum Loss Reduction, Subsample Percentage

obtained from the Turkish Statistical Institute (TUIK) and the Address-Based Population Registration System (ABPRS). Specifically, all-cause mortality data corresponding to the relevant years were used to populate the 'baseline incidence' parameter within AirQ+. This input is essential for the software to calculate the number of excess deaths attributable to exposure to $PM_{2.5}$. Although AirQ+ provides default relative risk (RR) values based on epidemiological studies conducted by the World Health Organization (WHO), the relevance of these values was enhanced by incorporating local mortality counts as population-specific baseline input. This approach enabled a more accurate representation of air pollution-related health outcomes in the study area.

Results and discussion

Real-world air quality prediction

ML models were created for various scenarios to determine the hourly pollutant levels in the study area. These models were based on data availability and variations in pollutant concentrations concerning different features. Two primary scenarios were considered:

1. In the first scenario, ML models were developed using only the ambient pollutant concentrations as input features.
2. In the second scenario, ML models were developed using emission and meteorological data as input features.

Scenario 1

As part of Scenario 1, the ambient pollutant concentrations for important air pollutants, including particulate matter with diameters less than 10 micrometers (PM_{10}), particulate matter with diameters less than 2.5 micrometers ($PM_{2.5}$), sulfur dioxide (SO_2), and carbon monoxide (CO), were utilized as input features for machine learning (ML) models. These pollutants significantly contribute to air pollution and affect public health and environmental quality. To obtain a robust and reliable model, the ML models were trained using a five-fold cross-validation technique, where the data is split into five subsets, trained on four, and validated on one iteratively. This helps reduce overfitting and variance (Samad et al. 2023). After training, the performance of the models was evaluated on a separate test dataset to assess how well they could predict new data. Hyperparameter tuning was performed to optimize the models further. Hyperparameters are parameters set before the learning process begins and can significantly impact the model's performance. The models can be fine-tuned by tuning these hyperparameters to achieve

better results. The model's performance was evaluated using specific metrics for each pollutant, such as RMSE, MAE, and R-squared. These metrics provide insights into how well the models predict pollutant concentrations and can help compare different models or approaches (Kumar and Pande 2023).

Various machine learning algorithms were evaluated after obtaining results from $PM_{2.5}$ and PM_{10} prediction models (Fig. 4). The Ridge model showed moderate performance for $PM_{2.5}$ with an R^2 value of 0.67 for test data. The SVR showed better accuracy with an R^2 value of 0.72. However, the RFR and ETR models showed significant overfitting tendencies, with much higher R^2 values for the training data (0.96 and 0.93) than the test data (0.79 for both). For PM_{10} predictions, the Ridge model performed slightly better, with an R^2 value of 0.4 for the test data, whereas the SVR model achieved an R^2 value of 0.44. The RFR and ETR models showed a considerable overfitting pattern, with substantially higher R^2 values for the training data (0.90 and 0.82) than the test data (0.45 and 0.46). These trends were consistent with the behaviors observed in the scenario analysis, where bagging methods like RFR and ETR tend to overfit the training data, leading to higher R^2 values than the test data (Liu et al. 2021). On the other hand, the Ridge and SVR models showed a more balanced performance, recording fewer R^2 values for the training data (Rybarczyk and Zalakeviciute 2018). Overall, the findings suggest that careful consideration of the model selection and parameter tuning is necessary to mitigate overfitting and enhance the generalization ability of the models (Zimmerman et al. 2018).

Upon analyzing the performance metrics for $PM_{2.5}$ and PM_{10} under Scenario-1, several noteworthy trends emerged among various regression models. In the $PM_{2.5}$ case, the Ridge regression model demonstrated moderate performance, with consistent MAE and RMSE values between the training and test sets. However, the SVR outperformed Ridge, demonstrating lower MAE and RMSE values and exhibiting better generalization from training to testing. The RFR also showed superior performance in the training set, boasting the lowest MAE and RMSE values. However, a discernible increase in both metrics on the test set was observed, suggesting potential overfitting.

Similarly, ETR and XGBoost performed well in the training set but showed signs of overfitting as MAE and RMSE values increased on the test set. It was found that Ridge regression performed consistently but with higher absolute error values in the PM_{10} scenario across training and testing. In contrast, the SVR demonstrated good predictive accuracy, with lower MAE and RMSE values, indicating effective generalization (2014). Notably, the RFR stood out as the top performer, achieving the lowest MAE and RMSE values in the training set (Islam et al. 2023). However, a notable increase in both metrics on the test set implied potential

Scenario 1

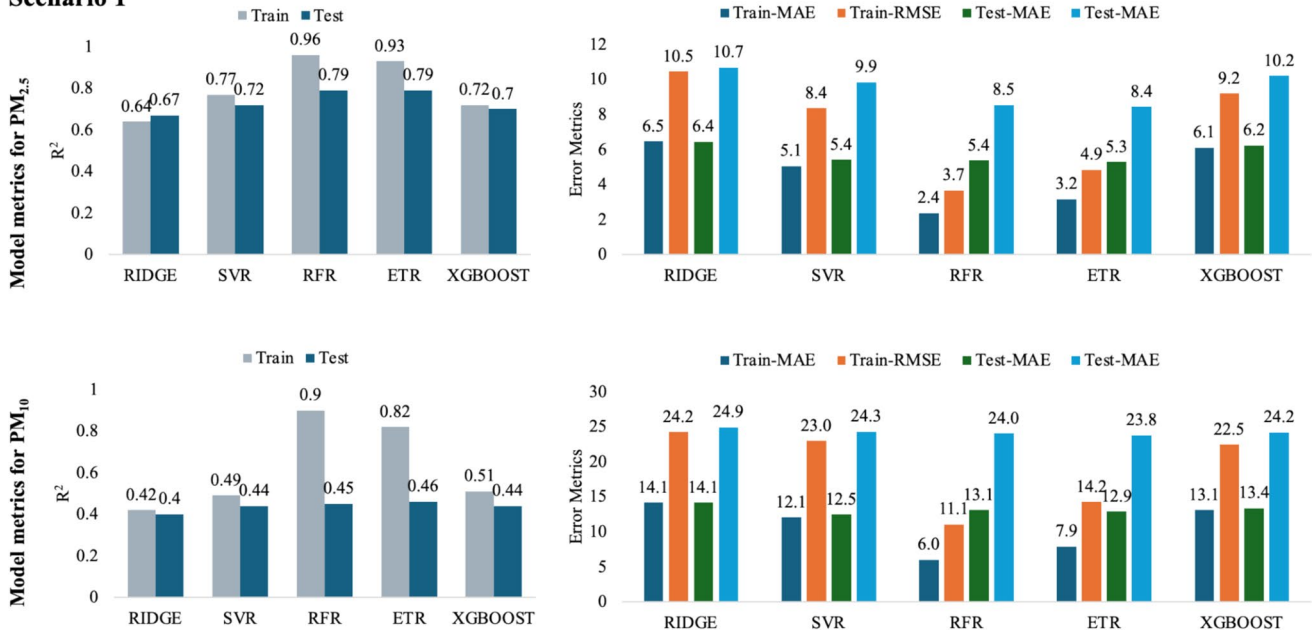


Fig. 4 Scenario 1 - error metrics for the prediction of PM_{2.5} and PM₁₀

overfitting, similar to the observations in the PM_{2.5} scenario. Similarly, ETR and XGBoost performed competitively in the training set but showed signs of overfitting on the test set (Datta et al. 2020). When evaluating regression models for both PM_{2.5} and PM₁₀ scenarios, it is essential to consider overfitting tendencies (Ma et al. 2022). While RFR consistently outperforms other models in the training set, it should be noted that it is mindful of addressing potential overfitting issues. SVR emerges as a robust choice, demonstrating effective generalization across both scenarios, while Ridge regression offers steady yet moderate performance. ETR and XGBoost show competitive results but require caution when mitigating overfitting for enhanced model reliability.

Scenario 2

Scenario 2 enhanced the prediction models for PM₁₀ and PM_{2.5} by incorporating meteorological data alongside ambient pollutant concentrations. This was done to gain a more comprehensive understanding of the factors influencing air quality. Five machine learning algorithms were again evaluated, and some interesting insights into their performance were discovered (Fig. 5). For PM_{2.5} predictions, the Ridge model showed moderate performance with an R² value of 0.68 on the test data. However, the SVR outperformed Ridge, achieving an R² value of 0.82, which indicates better accuracy. In reference to the study performed by Ma et al. (Ma et al. 2023), the authors observed that the ensemble models (RFR and ETR) demonstrated remarkable predictive power on the training data (R² = 0.97). Still, they exhibited

signs of overfitting, as reflected in substantially higher R² values for the training data than the test data. XGBOOST also showed competitive performance, but caution is needed to address potential overfitting, such as the ensemble models (Gokul et al. 2023). For PM₁₀ predictions, the Ridge model performed slightly better with an R² value of 0.47 for the train data, while SVR achieved an R² value of 0.58. Like PM_{2.5}, the ensemble models (RFR and ETR) displayed overfitting tendencies, with higher R² values (0.92 and 0.89) on the training data than the test data. XGBOOST also demonstrated performance but with a potential risk of overfitting. It is evident from Krylova and Okhrin's study (Krylova and Okhrin 2021) that including meteorological data has improved the model performance when compared between the two scenarios. Ridge regression and SVR consistently demonstrated more balanced results, indicating better generalization abilities. On the other hand, ensemble models like RFR and ETR exhibited overfitting tendencies, emphasizing the importance of careful model selection and hyperparameter tuning.

In the case of PM₁₀, Ridge regression showed consistent yet moderate performance across both training and test sets, with higher MAE values compared to the PM_{2.5} scenario. On the other hand, SVR demonstrated effective generalization with lower MAE and RMSE values, showcasing its robust predictive accuracy (Lei et al. 2023). The RFR model was a top performer in the training set for PM₁₀ and PM_{2.5}, with the lowest MAE and RMSE values (Lei et al. 2023). However, an increase in both metrics on the test set was observed, indicating potential overfitting.

Scenario 2

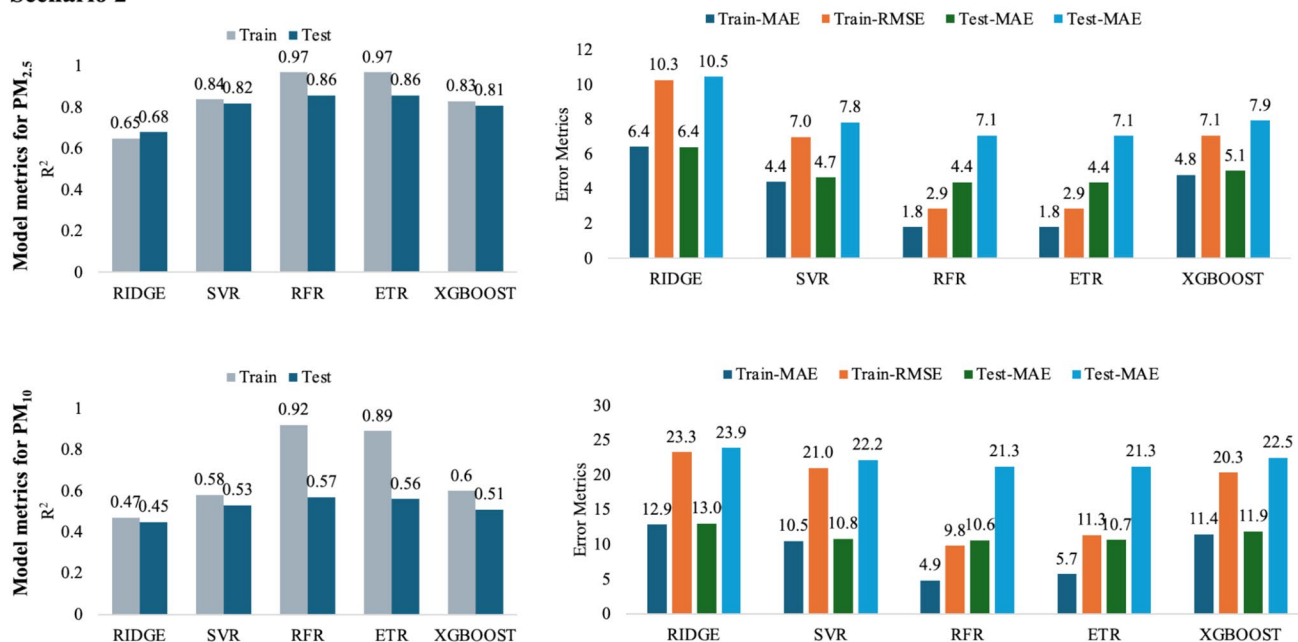


Fig. 5 Scenario 2 - error metrics for the prediction of PM_{2.5} and PM₁₀

Similar trends were observed with ETR and XGBoost, demonstrating competitive performance in the training set but revealing signs of overfitting on the test set. In contrast to Scenario 1, where only ambient pollutant concentrations was utilized, incorporating meteorological data in Scenario 2 has impacted model performance. Ridge regression consistently displayed steady yet moderate performance, while SVR emerged as a reliable choice, demonstrating effective generalization across both scenarios. The overfitting tendencies observed in random forest regression consistently outperform other models in the training set. Caution must be taken when addressing potential overfitting issues to enhance model reliability. In contrast, SVR exhibits strong generalization, making it a reliable choice in both scenarios.

In summary, integrating meteorological data alongside ambient pollutant concentrations in Scenario 2 has provided a more nuanced understanding of air quality prediction (Imam et al. 2024). While some models exhibit overfitting, carefully considering model selection and hyperparameter tuning can enhance the models' overall reliability and generalization ability for both PM₁₀ and PM_{2.5} scenarios. Additionally, studies have shown that meteorological variables significantly impact the prediction of air pollutants (Wu et al. 2024; Tao et al. 2023). These variables, which include temperature, humidity, wind speed, and direction, affect the dispersion and concentration of pollutants in the atmosphere. Understanding the relationship between meteorological factors and air quality is crucial in developing

effective strategies to mitigate the adverse effects of air pollution on human health and the environment.

Subsequent 24-hour prediction

This study conducted an extensive examination to evaluate the predictive capabilities of five distinct machine learning models. The dataset used in the study spanned four years, from 01.01.2019 to 31.12.2022, and it included PM_{2.5}, PM₁₀, and various meteorological parameters. Rather than internally generated test data, real-world data assesses the models' effectiveness, making the evaluation more reliable and effective. The models were trained on historical datasets, including data up to 31.12.2022, to ensure that they were equipped to predict PM_{2.5} and PM₁₀ concentrations for the subsequent 24 hours. Then, the models' ability is thoroughly assessed to make such predictions using actual pollutant and meteorological data for the next 24 hours, starting from 01.01.2023. To improve air quality, the study aimed to evaluate which machine learning models were most effective at predicting pollutant concentrations based on real-world data. The models were evaluated based on their accuracy, precision, sensitivity, and specificity. Using real-world data and assessing the models' effectiveness thoroughly generated valuable insights to help develop effective pollution control and prevention strategies (Huang et al. 2021).

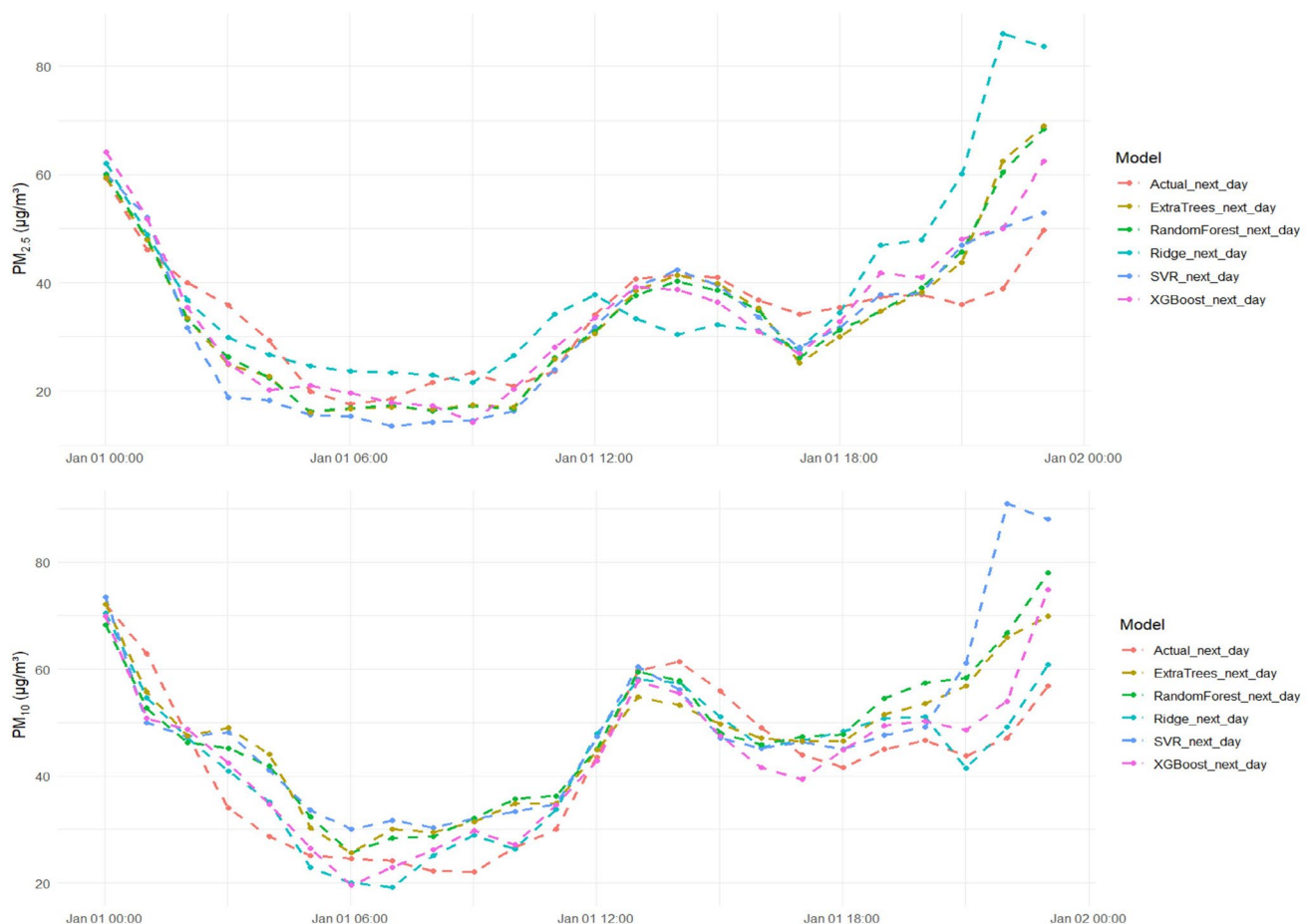
Performance metrics, as shown in Table 2, including R², MAE, and RMSE, were used to assess the strengths and weaknesses of each model. After an analysis, a visual

Table 2 Performance metrics for the prediction of subsequent 24-hour

ML Models	PM _{2.5}			PM ₁₀		
	R ²	MAE	RMSE	R ²	MAE	RMSE
Ridge	0.44	9.20	14.11	0.91	3.95	4.48
SVR	0.83	5.03	6.62	0.51	9.08	13.35
RFR	0.77	5.28	7.41	0.71	7.72	9.45
ETR	0.76	5.23	7.73	0.75	7.02	8.58
XGBOOST	0.80	5.25	6.43	0.81	5.16	6.47

representation of the actual values and model results on a timeline is provided in Fig. 6, which offers a comprehensive view of the predictive performance over the evaluated period. The graphical representation in Fig. 6 improves the understanding of how effectively each model captures the dynamics of PM_{2.5} and PM₁₀ concentrations, providing a valuable complement to the quantitative metrics presented in Table 2. The Ridge Regression model showed a moderate R² value of 0.44 for PM_{2.5} prediction and an impressive R² of 0.91 for PM₁₀ prediction. The lower R² value of 0.44 for Ridge Regression in PM_{2.5} prediction, compared to higher R² values observed in other models (ranging from 0.7 to

0.9), is mainly due to Ridge Regression's limitations in capturing complex, nonlinear relationships in the data. Ridge Regression is a linear model that effectively prevents overfitting in high-dimensional datasets but lacks the flexibility to model the nonlinear interactions that might be present in the air quality data, especially under varying meteorological conditions. In contrast, models such as Random Forest, Extra Trees, and XGBoost, which use decision trees and ensemble learning techniques, are more adept at handling these nonlinear relationships, achieving higher predictive accuracy and R² values. The difference is probably because of multicollinearity and complex interactions between

**Fig. 6** Comparison of PM_{2.5} and PM₁₀ across different machine learning models (MLs) and actual concentration levels

predictors that Ridge Regression may not wholly address despite its regularization mechanism. Nonlinear models are better at exploiting these relationships, resulting in more accurate $PM_{2.5}$ predictions. These factors explain why Ridge Regression shows a lower R^2 , especially for $PM_{2.5}$ prediction, but performs better in more straightforward, linear scenarios like PM_{10} forecasting. The SVR model also performed well, particularly in $PM_{2.5}$ prediction, with an R^2 of 0.83. However, it faced challenges in accurately forecasting PM_{10} . The RFR and ETR models showed robust overall performance across $PM_{2.5}$ and PM_{10} prediction tasks. The ETR model outperformed other models in PM_{10} forecasting with an R^2 of 0.75. The XGBoost model demonstrated competitive results for both $PM_{2.5}$ and PM_{10} , with R^2 values of 0.80 and 0.81, respectively. XGBoost delivers exceptional performance because it can capture intricate, nonlinear relationships within the data. It also adeptly manages missing data. Moreover, its parallel processing capability enhances efficiency, allowing the model to swiftly and accurately handle real-world datasets.

The use of real-world data in this study enhances the practical relevance of the findings. The reported results offer valuable insights into the complex dynamics of air quality prediction and highlight the importance of carefully weighing each model's strengths and limitations. The focus on forecasting the subsequent 24 hours underscores the models' effectiveness in capturing short-term trends in pollutant concentrations and meteorological conditions, as evidenced by the reported results. These findings provide a

foundation for future research and development of machine learning models for air quality prediction.

Health impact assessment-AirQ⁺

The AirQ⁺ analysis conducted for Aksaray from 2019 to 2022 examined the estimated number of health issues per 100,000 people due to $PM_{2.5}$ exposure (Fig. 7). In 2019, there were an estimated 52 cases per 100,000 people, with a lower estimate of 40 and an upper estimate of 58. In 2020, the estimated number increased to 59 cases per 100,000, with a lower estimate of 45 and an upper estimate of 66. The year 2021 showed a further increase, with an estimated 95 cases per 100,000, ranging from 73 to 106. Finally, in 2022, there were an estimated 83 cases per 100,000, with a lower estimate of 64 and an upper estimate of 92. These numbers indicate a concerning trend of increasing health issues attributable to $PM_{2.5}$ exposure over the years in Aksaray (2022). The findings from the AirQ⁺ analysis are consistent with previous research highlighting the detrimental health effects of particulate matter ($PM_{2.5}$) pollution on the population (Rovira et al. 2020). Studies have shown that exposure to $PM_{2.5}$ is associated with various health issues, including respiratory problems, cardiovascular diseases, and even cancer (Sharma et al. 2020). The estimates of attributable cases per 100,000 people provide a quantitative measure of the impact of $PM_{2.5}$ exposure on public health in Aksaray. The increasing trend in the estimated number of cases over the years underscores the urgent need for interventions to mitigate air

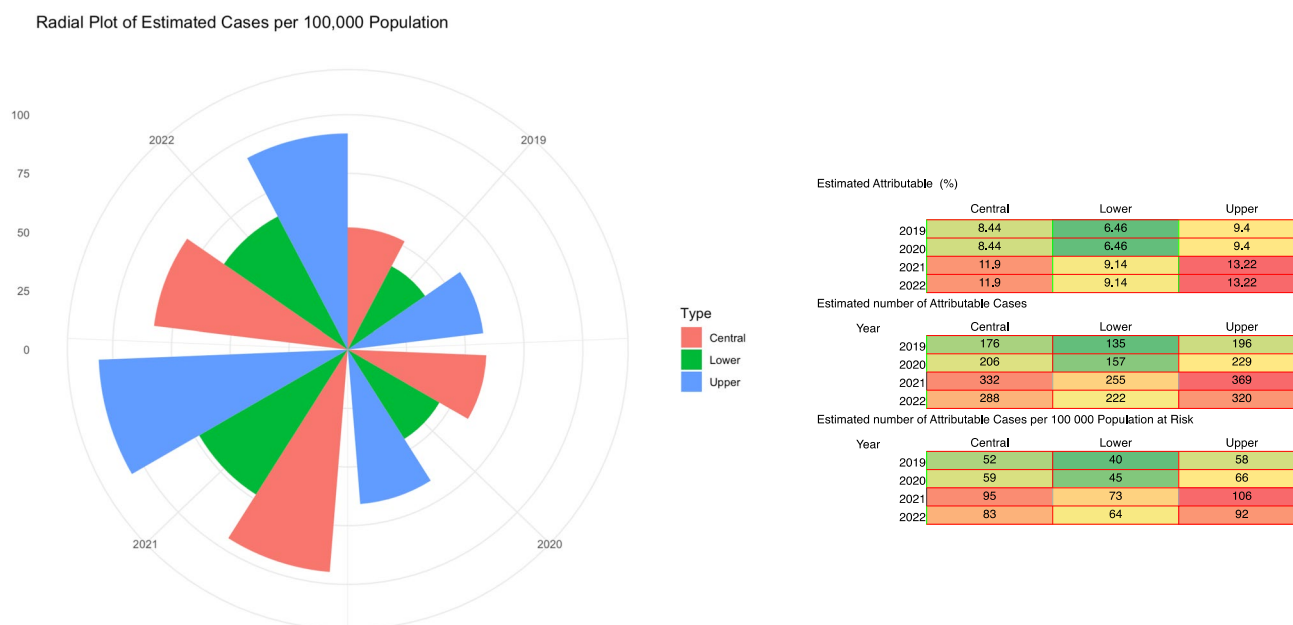


Fig. 7 AirQ⁺ Radial plot of Estimated number of attributable cases per 100,000 population at risk and results of AirQ⁺

pollution and protect the health of the residents in the region (Guarnieri and Balmes 2014).

Additionally, the results of the AirQ+ analysis for Aksaray emphasize the importance of considering regional air pollution levels and long-term exposure to air pollutants when assessing the health effects of $PM_{2.5}$. The study's focus on $PM_{2.5}$ aligns with existing research that identifies this pollutant as a significant contributor to adverse health outcomes. While $PM_{2.5}$ is a well-studied pollutant with known health impacts, it is crucial to acknowledge the role of other pollutants like nitrogen oxides and carbon monoxide, which can also pose health risks and often correlate with $PM_{2.5}$ levels. Therefore, comprehensive strategies addressing multiple pollutants are essential to effectively safeguard public health in areas like Aksaray experiencing concerning levels of air pollution (2024).

Given the increasing burden of health issues attributable to $PM_{2.5}$ exposure in Aksaray, interventions aimed at reducing air pollution levels and protecting the population from harmful particulate matter become imperative. Strategies such as enhancing emission controls, promoting cleaner energy sources, implementing urban planning measures to reduce traffic congestion, and raising public awareness about the health risks of air pollution are crucial in mitigating the adverse effects of $PM_{2.5}$. Additionally, policies focusing on improving air quality standards, monitoring pollution levels, and enforcing regulatory measures can significantly safeguard public health and reduce the number of cases attributed to $PM_{2.5}$ exposure in the region. Collaborative efforts involving government agencies, environmental organizations, healthcare providers, and the community are essential to address the escalating public health challenges associated with air pollution in Aksaray and ensure a healthier environment for all residents (Koçak and Balcılar 2024).

In conclusion, the results of the AirQ+ analysis for Aksaray from 2019 to 2022 demonstrate a concerning increase in the estimated number of attributable cases of health issues per 100,000 people due to $PM_{2.5}$ exposure. These findings emphasize the urgent need for targeted interventions and comprehensive policies to mitigate air pollution levels, reduce the health risks associated with particulate matter pollution, and protect the well-being of the residents in the region. By implementing effective strategies to improve air quality and promote a healthier environment, Aksaray can work towards reducing the burden of preventable health issues linked to $PM_{2.5}$ exposure and enhance the overall quality of life for its population.

Limitations and considerations

The present study focused on integrating emission and meteorological data, potentially overlooking influential factors. The absence of a more diverse set of inputs, such as

land-use patterns, demographic information, or industrial emissions, may limit the models'ability to capture the full spectrum of contributors to air quality. The introduction of vehicle ambient pollutant concentrations, while aiming to enhance model comprehensiveness, raises concerns about potential overfitting, especially with increased model complexity. Striking a balance between incorporating diverse inputs and managing overfitting remains a nuanced challenge. The models may face challenges in effectively capturing emissions' temporal and spatial variations, especially with different sources contributing at varying rates. Ensuring the models can adapt to these dynamic patterns is crucial for accurate predictions. Integrating a broader range of data types introduces complexities in handling diverse temporal and spatial resolutions. Harmonizing datasets with varying scales requires careful consideration to maintain the integrity of the predictive models. Future research should explore the inclusion of additional factors such as land-use patterns, demographic characteristics, and industrial emissions to create a more comprehensive input set. This can provide a holistic understanding of the multifaceted influences on air quality. Developing models that dynamically adapt to input types and source changes, allowing for seamless integration of diverse data, can enhance the models'flexibility and responsiveness to evolving environmental conditions. Exploring advanced feature engineering methods can help extract meaningful patterns from diverse inputs, mitigating the risk of overfitting. Techniques like dimensionality reduction and feature selection can enhance model interpretability and generalizability.

Future work should incorporate uncertainty quantification measures tailored explicitly to diverse inputs. Understanding the confidence levels of various input types contributes to a more transparent and reliable predictive framework. Collaborating with stakeholders to access a broader range of data types can enrich the input diversity. Establishing partnerships with relevant entities, including urban planning departments and industrial regulatory bodies, can facilitate the integration of diverse and valuable datasets. By addressing these limitations and pursuing avenues for incorporating a more diverse set of inputs, the predictive models can evolve into robust tools for comprehensive air quality assessment, aiding in developing effective pollution control strategies. The study had another limitation, as it only recorded the total number of deaths in the studied region without specifying the diseases that caused them. This lack of specificity made it impossible to estimate the contribution of different types of diseases, such as acute lower respiratory tract disease (ALRI), chronic obstructive pulmonary disease (COPD), and ischemic heart disease (IHD). Despite these limitations, the study has provided an essential record of the impact of air quality on health. It is expected that future studies with expanded data sets can produce even more reliable results.

Conclusions

This study examined the use of machine learning models to predict hourly pollutant concentrations and assess their potential health impacts in a specific urban area. Two scenarios were analyzed: one focused solely on ambient pollutant concentrations, while the other combined ambient pollutant concentrations with meteorological variables. The study employed several machine learning algorithms, including Ridge Regression, Support Vector Regression (SVR), Random Forest Regressor (RFR), Extra Trees Regressor (ETR), and XGBoost.

In the first scenario, Ridge Regression and SVR showed strong predictive performance, particularly for $PM_{2.5}$. However, ensemble-based models like RFR and ETR exhibited signs of overfitting, highlighting the need for careful model selection and hyperparameter tuning when using these approaches. By incorporating meteorological variables in the second scenario, the models gained a more profound understanding of air quality dynamics, emphasizing the complex interactions between weather patterns and pollutant behavior.

To further evaluate the robustness of the models, a 24-hour prediction experiment was conducted using real-world data collected over four years. SVR proved particularly effective in forecasting $PM_{2.5}$ concentrations, while ETR excelled in predicting PM_{10} levels. These findings underscore the importance of aligning model selection with the characteristics of the pollutants and the timeframes involved. The application of real-world testing demonstrated the models' practical utility for short-term air quality management.

Finally, the study incorporated the AirQ+ model to estimate the long-term health impacts associated with $PM_{2.5}$ exposure. The results indicated an average of 250 attributable deaths per year, with annual values ranging from 135 to 369. This underscores the urgent need for public health interventions aimed at reducing exposure to air pollution. Overall, the study provides insights that are not only relevant for predictive modeling but also crucial for policy development and public health planning.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s12145-025-01941-7>.

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Author contributions E.K. contributed to writing the original draft, reviewing and editing, conceptualization, methodology, and software

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Data availability The datasets and materials used and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval Not applicable.

Clinical trial number Not applicable.

Consent to Participate Not applicable.

Consent for Publication Not applicable.

Conflict of interests The authors declare no competing interests.

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